

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 25: Probability

Topic-wise short notes with solved examples

Probability is a number between 0 and 1 that measures how likely an event is. This chapter covers the probability of an event when all the outcomes are equally likely.

1 Basic terms

- A **random experiment** is one whose result cannot be predicted in advance, such as tossing a coin, throwing a die or drawing a card.
- Each possible result is an **outcome**. The set of all possible outcomes is the **sample space**, S .
- An **event** E is any set of outcomes. The outcomes in it are the **favourable outcomes**.
- Outcomes are **equally likely** when none is expected to occur more often than another.

2 Probability of an event

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} = \frac{n(E)}{n(S)}$$

- $0 \leq P(E) \leq 1$.
- An **impossible event** has probability 0; a **sure (certain) event** has probability 1.
- The event "not E " is the **complementary event** $\bar{E}$:

$$P(E) + P(\bar{E}) = 1$$

Example 2.1 A die is thrown once. Find the probability of getting (i) an even number (ii) a multiple of 3 (iii) a number less than 1 (iv) a number not greater than 6.

SOLUTION

$S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$

(i) $\{2, 4, 6\}$: $P = \frac{3}{6} = \frac{1}{2}$ (ii) $\{3, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$

(iii) No such number: $P = 0$ (iv) All six numbers: $P = 1$

(i) $\frac{1}{2}$ (ii) $\frac{1}{3}$ (iii) 0 (iv) 1

3 Coins and dice

Experiment	Number of outcomes	Sample space
One coin	2	H, T
Two coins	4	HH, HT, TH, TT
Three coins	8	HHH, HHT, HTH, THH, HTT, THT, TTH, TTT
One die	6	1, 2, 3, 4, 5, 6
Two dice	36	Ordered pairs (1, 1) to (6, 6)

Example 3.1 Three coins are tossed together. Find the probability of getting (i) exactly two heads (ii) at least one tail.

SOLUTION

$$n(S) = 8$$

$$(i) \text{ HHT, HTH, THH: } P = \frac{3}{8}$$

$$(ii) \text{ "At least one tail" is the complement of "no tail" (HHH): } P = 1 - \frac{1}{8}$$

$$(i) \frac{3}{8} \quad (ii) \frac{7}{8}$$

Example 3.2 Two dice are thrown together. Find the probability that (i) the sum is 7 (ii) the two numbers are equal (iii) the sum is more than 10.

SOLUTION

$$n(S) = 36$$

$$(i) (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1): P = \frac{6}{36} = \frac{1}{6}$$

$$(ii) (1, 1), (2, 2), \dots, (6, 6): P = \frac{6}{36} = \frac{1}{6}$$

$$(iii) (5, 6), (6, 5), (6, 6): P = \frac{3}{36} = \frac{1}{12}$$

$$(i) \frac{1}{6} \quad (ii) \frac{1}{6} \quad (iii) \frac{1}{12}$$

4 Playing cards

- A pack has **52 cards**: 4 suits of 13 cards each.
- Spades and clubs are **black** (26 cards); hearts and diamonds are **red** (26 cards).
- Each suit has an ace, the numbers 2 to 10, a jack, a queen and a king.
- **Face cards**: jack, queen and king. There are 12 face cards, 6 red and 6 black.

Example 4.1 A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is (i) an ace (ii) a black king (iii) a face card (iv) neither a heart nor a king.

SOLUTION

$$(i) 4 \text{ aces: } P = \frac{4}{52} = \frac{1}{13} \quad (ii) 2 \text{ black kings: } P = \frac{2}{52} = \frac{1}{26}$$

$$(iii) 12 \text{ face cards: } P = \frac{12}{52} = \frac{3}{13}$$

$$(iv) \text{ Hearts or kings: } 13 \text{ hearts} + 3 \text{ other kings} = 16 \text{ cards, so } 52 - 16 = 36 \text{ cards: } P = \frac{36}{52} = \frac{9}{13}$$

$$(i) \frac{1}{13} \quad (ii) \frac{1}{26} \quad (iii) \frac{3}{13} \quad (iv) \frac{9}{13}$$

5 Balls, numbered cards and letters

Example 5.1 A bag contains 4 red, 6 blue and 5 green balls. One ball is drawn at random. Find the probability that it is (i) blue (ii) not green (iii) red or green.

SOLUTION

Total = 15 balls

$$(i) P = \frac{6}{15} = \frac{2}{5} \quad (ii) P = \frac{10}{15} = \frac{2}{3} \quad (iii) P = \frac{9}{15} = \frac{3}{5}$$

$$(i) \frac{2}{5} \quad (ii) \frac{2}{3} \quad (iii) \frac{3}{5}$$

Example 5.2 Cards numbered 1 to 25 are placed in a box and one is drawn at random. Find the probability that the number is (i) prime (ii) a perfect square (iii) divisible by both 2 and 3.

SOLUTION

$$(i) \text{ Primes: } 2, 3, 5, 7, 11, 13, 17, 19, 23, \text{ which is 9 numbers: } P = \frac{9}{25}$$

$$(ii) 1, 4, 9, 16, 25: P = \frac{5}{25} = \frac{1}{5}$$

$$(iii) \text{ Multiples of 6: } 6, 12, 18, 24: P = \frac{4}{25}$$

$$(i) \frac{9}{25} \quad (ii) \frac{1}{5} \quad (iii) \frac{4}{25}$$

Example 5.3 A letter is chosen at random from the word PROBABILITY. Find the probability that it is (i) a vowel (ii) the letter B.

SOLUTION

The word has 11 letters: P, R, O, B, A, B, I, L, I, T, Y.

$$(i) \text{ Vowels: } O, A, I, I, \text{ which is 4 letters: } P = \frac{4}{11}$$

$$(ii) \text{ B appears twice: } P = \frac{2}{11}$$

$$(i) \frac{4}{11} \quad (ii) \frac{2}{11}$$

6 Finding an unknown number

Example 6.1 A bag contains 12 balls, of which x are white. (i) Write the probability of drawing a white ball. (ii) If 6 more white balls are added, the probability of drawing a white ball doubles. Find x .

SOLUTION

(i) $P = \frac{x}{12}$

(ii) Now there are $x + 6$ white balls out of 18: $\frac{x + 6}{18} = 2 \times \frac{x}{12}$

$$\frac{x + 6}{18} = \frac{x}{6}, \text{ so } x + 6 = 3x$$

$$x = 3$$

7 Common mistakes

- Writing a probability greater than 1.
- Treating (3, 4) and (4, 3) as the same outcome for two dice.
- Counting 1 as a prime number.
- Counting the king of hearts twice in "a heart or a king".
- Leaving the answer unsimplified.